

A Network and Its Reflection: What True, False, and Noise Self-Model Broadcasts Do and Do Not Change in an Evolving Multi-Agent Network

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WWW home page: <https://github.com/Ivan825/Project-ONE>

Abstract. Many networks respond to measurements of their own state: markets to indices, organizations to metrics, platforms to trending signals. We present a controlled laboratory for this reflexive loop: an evolving agent population forms an adaptive network whose macrostate is compressed into a self-model and broadcast back — accurately, distorted, or as matched-range noise. Across 927 deterministic paired-seed runs: measurement without broadcast is causally inert. A natural self-fulfilling-prophecy statistic registers strong apparent attraction toward false and replayed self-models — but a passive counterfactual, scoring untreated twin trajectories against the same reference signals, fully accounts for it: a caution for empirical claims of performativity. What broadcasts demonstrably change is behavior and structure, and which lies come true is set jointly by the lie and the response regime. Finally, heritable attention to the broadcast declines across generations, fastest under unreliable, high-impact feedback: populations fed false self-models evolve to ignore their observer.

Keywords: adaptive networks, agent-based modeling, reflexivity, self-fulfilling dynamics, performative prediction, evolution of trust

1 Introduction

Adaptive networks — graphs whose topology coevolves with the dynamical states of their nodes — are a central object of network science [1, 2]. A distinctive subclass has received far less controlled attention: networks that respond to *descriptions of themselves*. Financial markets move on published indices computed from their own trades [3]; universities restructure around rankings derived from their own behavior [4]; recommendation platforms amplify items *because* an internal measurement declared them trending. In each case a macroscopic measurement is compressed, published, and re-enters the system as an input to microscopic decisions — a loop Merton called the self-fulfilling prophecy [5] and whose measure-distorting variant is known as Goodhart’s law [6].

Empirical study of this loop is confounded: in real systems one can rarely withhold the measurement, falsify it in controlled directions, or replay history

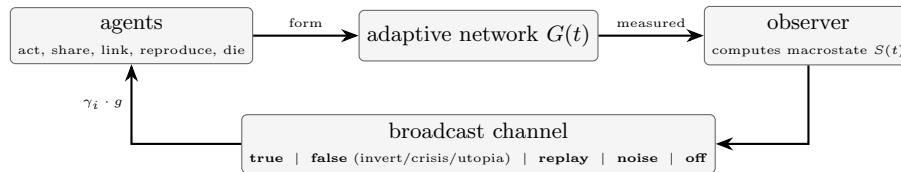


Fig. 1. The reflexive loop and its corruption points. Local actions generate the network; the observer compresses it into a self-model; the broadcast channel returns it truthfully, distorted, replayed from another run, as noise, or not at all. Action-propensity responses are scaled by heritable sensitivity γ_i and global gain g ; the hub-targeted pruning probability is scaled by γ_i alone.

under identical randomness. We therefore built a minimal simulated world in which the loop can be opened, closed, and corrupted at will, where every run is exactly reproducible from a configuration and seed, and where agents are deliberately simple — transparent trait vectors rather than learned policies — so every pathway from broadcast to response is inspectable.

Our contributions are: (1) an open, deterministic experimental platform coupling an evolving agent ecology, an adaptive interaction network, a global observer, and a controllable broadcast channel; (2) a six-condition causal design — verifying observer-side-effect invariance and isolating the effect of broadcasting the measurement, with matched-range noise and replayed-self-model controls — together with a *passive-counterfactual methodology* for reflexivity metrics: a natural “story pull” statistic registers strong apparent steering toward false and replayed self-models, and we show it is fully accounted for by intrinsic dynamics measured on untreated twin trajectories, a direct caution for empirical tests of performative prediction; (3) evidence for what broadcasts *do* change: truthful feedback doubles costly cooperation, distorted broadcasts mobilize or pacify depending on the lie–response pairing (reversing under conformist responses), and broadcasts densify and decentralize the network; and (4) we demonstrate, in an ecological agent-based setting, that *attention to a collective self-model is itself a heritable, evolvable trait* whose population mean declines most strongly under unreliable, high-impact self-models. Figure 1 summarizes the design.

2 Related Work

Adaptive and temporal networks. Coevolution of state and topology is surveyed in [1, 2]; our model belongs to this family, with births, deaths and rewiring driven by agent decisions. Robustness of networks to random and targeted node removal is classical [7]; we use targeted hub removal as a standardized perturbation. **Global coupling in physics.** Globally coupled oscillators and maps show that a mean field fed back to units reshapes collective dynamics [8]; here the “mean field” is a multidimensional, potentially *false* description, and units are born, die, and evolve. **Reflexivity and reactive measurement.** Self-fulfilling prophecy [5], reflexivity in markets [3], and the reactivity of rankings

and metrics [4, 6] motivate our false-broadcast conditions; we contribute a platform where these effects are dosable and ablatable. **Performative prediction and endogenous signals.** Predictions that alter the distribution they forecast are formalized as performative prediction [19], including networked multi-agent settings [20] and populations that decide whether to trust predictions of collective outcomes [21]; self-fulfilling signals of an endogenous network state arise in congestion games [22]. Our contribution to this line is a testbed in which apparent performativity can be checked against passive counterfactuals — a check our own headline metric initially failed (Sect. 5.2) — plus population turnover and a heritable receiver-sensitivity trait absent from these models. **Signaling and the evolution of trust.** In sender–receiver games, receivers can evolve to ignore unreliable signals [9]. Our distrust result is consonant with that theory but arises in an open ecological population where the signal is *self-referential* and attention to it is a continuous heritable trait in an energy economy, not a payoff matrix. **Agent-based modeling.** Agent-based studies of collective opinion and cooperation are a mainstay of this community [16, 17]; our methodology follows the generative ABM tradition [10, 11], and the engine uses ideas from Mesa [12] and NetworkX [13]. Multi-agent reinforcement learning treats global state as an *architectural* aid [14]; we treat it as an *experimental variable* with controlled veracity. Populations of language-model agents are increasingly used to study emergent social behavior [15]; shared dashboards describing such populations’ collective state are a natural coordination design there, and the present rule-based setting provides causally clean baselines for it.

3 The Model

3.1 Agents and ecology

Each agent i carries a heritable trait vector $\theta_i = (\text{coop}, \text{expl}, \text{risk}, \text{soc}, \text{share}, \gamma_i) \in [0, 1]^6$, where γ_i is its *global sensitivity* — how strongly broadcasts modulate its behavior. Agents hold energy; each step they pay metabolism and choose one action — harvest, share (costly transfer to a neighbor), connect, prune, reproduce, or idle — sampled with probabilities proportional to trait- and state-dependent propensities. The resource pool regenerates logistically, giving density-dependent competition. Reproduction requires an energy threshold and copies traits with Gaussian mutation ($\sigma = 0.05$); energy exhaustion or age causes death. Founding ages are staggered to avoid cohort artifacts.

3.2 Observer and broadcast conditions

Every $\Delta t = 10$ steps the observer computes the macrostate $S(t)$: population, fragmentation (1 minus largest-component share), degree- and betweenness-concentration, Freeman centralization [18], per-capita costly-helping rate, energy Gini, and turnover, plus mean trait values (logged, never broadcast). The broadcast $b(t)$ carries five of these components, each normalized to $[0, 1]$: *fragmentation*, *degree concentration* (top-5% degree share; distinct from Freeman centralization), *cooperation*, *inequality*, and *turnover*. Six conditions define what agents

Table 1. Model parameters (defaults; campaign overrides noted in the text).

initial population	150	observer interval Δt	10 steps
population cap	1200	feedback gain g	0.8 (swept 0.2–1.6)
initial edges per agent	3 (random)	mutation σ	0.05
initial energy	20	initial traits	mean 0.5, s.d. 0.15
resource capacity K	6000	reproduction threshold / cost	30 / 16
base regeneration	60/step	min. reproduction age	15 steps
regeneration rate r	0.12	lifespan (mean $\pm$ s.d.)	400 $\pm$ 80 steps
harvest cap	2.5/step	action cost / share amount	0.25 / 3.0
metabolism	0.8/step	run length	3000–4000 steps
shock time t_{shock}	2000 (1500 in the $2\times$ scale check)		

receive (Table 2). Distorted broadcasts come in three modes: *invert* ($b_k = 1 - s_k$ componentwise), *crisis* (fixed alarming values: fragmentation 0.9, degree concentration 0.9, cooperation 0.1, inequality 0.9, turnover 0.9), and *utopia* (fixed reassuring values: 0.05, 0.2, 0.9, 0.1, 0.2).

3.3 Response model

Each step, agent i selects one action a with probability proportional to

$$w_i(a, t) = w_a(\theta_i, x_i(t)) + \gamma_i g f_a(b(t)), \quad (1)$$

where $w_a(\theta_i, x_i)$ is the trait- and local-state-dependent base propensity, g is the global feedback gain, and f_a is the response rule. In the primary *corrective* regime agents repair reported deficits: $f_{\text{connect}} = \max(0, b_{\text{frag}} - 0.3)$, $f_{\text{share}} = \max(0, 0.5 - b_{\text{coop}}) + 0.5 \max(0, b_{\text{ineq}} - 0.5)$, $f_{\text{harvest}} = \max(0, b_{\text{turn}} - 0.5)$, and $f_{\text{prune}} = 0.3 \max(0, b_{\text{cent}} - 0.6)$. In the *conformist* regime agents imitate the reported norm instead: $f_{\text{share}} = b_{\text{coop}}$, $f_{\text{prune}} = 0.5 b_{\text{frag}}$, $f_{\text{connect}} = \max(0, 0.6 - b_{\text{frag}})$, and $f_{\text{harvest}} = b_{\text{ineq}}$. Two further mechanisms act on *target selection* rather than action choice, in both regimes: while reported degree concentration exceeds 0.6, a pruning agent cuts its highest-degree neighbor (ties by id) with probability γ_i — note g scales action propensities only, not this target-selection probability — else a uniform neighbor; and new links attach to a neighbor-of-a-neighbor with probability 0.7 when available (else a uniform non-neighbor) — an explicit triadic-closure tendency contributing to the clustering in Sect. 3.4. All rule constants are fixed across all experiments; Table 1 lists the model parameters.

3.4 Emergent network structure

The emergent networks are dense, clustered, and heterogeneous: at $t = 2,000$ the giant component has $L \approx 1.9$ – 2.5 , clustering 1.8 – $2.4\times$ a matched Erdős–Rényi null, and degree coefficient of variation 0.6 – 0.75 . Against the stricter degree-preserving (configuration-model) null most of the clustering excess is accounted for by the degree sequence (ratios 1.0 – 1.6), so we make no small-world claim:

Table 2. Experimental conditions. All start from matched parameter distributions with paired random seeds.

Condition	Observer	Agents receive	Role
A — Local only	off	nothing	baseline
B — Observed, blind	on	nothing	measurement control
C — True feedback	on	accurate $S(t)$	main treatment
F — False feedback	on	distorted $S(t)$	reflexivity probe
N — Noise feedback	on	uniform random, matched ranges	stimulation control
R — Replayed self-model	on	another run’s genuine $S(t)$	structure control

feedback acts on a small, dense, degree-heterogeneous clustered graph rather than a homogeneous mixing population.

3.5 Reproducibility

The simulation is deterministic given (configuration, seed): one PRNG drives all behavioral choices, a separate dedicated stream generates broadcast signals (so conditions differ only in signal *content*, never in RNG consumption), agents act in sorted order, and a state hash certifies replay; identical hashes were verified across operating systems and machines. All code, configurations, and analysis scripts are public.

4 Experimental Design

The standardized shock removes, in a single step at $t_{\text{shock}} = 2000$ (except in the $2\times$ scale check: $t_{\text{shock}} = 1500$ in a 3,000-step run), the highest-degree living agents (ties broken by agent id): a fixed 10 hubs in the mild variant, the top $\max(1, \lfloor 0.4 |V_{\text{live}}| \rfloor)$ in the harsh variant. Four outcome *families* were fixed before the first campaign: (1) post-shock recovery time — the first $t > t_{\text{shock}}$ at which the living population regains 90% of its pre-shock baseline, defined as its mean over $[t_{\text{shock}} - 500, t_{\text{shock}}]$; (2) mean post-shock fragmentation; (3) per-capita costly-helping rate; (4) self-model correspondence, operationalized as *story pull*: with $s_k(t)$ the value of macrostate component k at observer tick t , $b_k(t)$ the broadcast value, and $\varepsilon = 0.02$,

$$P = \left\langle \text{sign}(b_k(t) - s_k(t)) (s_k(t+1) - s_k(t)) \right\rangle_{(t,k) : |b_k(t) - s_k(t)| > \varepsilon}, \quad (2)$$

averaged with equal weight over all qualifying (tick, component) pairs across the five broadcast components; $P > 0$ means the macrostate moves toward the reference signal where the two differ. The threshold ε makes P well defined under false and noise broadcasts (and undefined for C, whose broadcast equals the current state). Eq. 2 is a directional refinement, introduced after the first campaign, of the pre-specified correspondence outcome, whose original distance-based form is degenerate under truthful feedback; both give the same raw F-vs-N

Table 3. Main paired comparisons (Wilcoxon signed-rank on per-seed differences; matched-pairs rank-biserial r ; bootstrap 95% CI of the median difference). FL = flagship (50 pairs), HS = harsh shock (30 pairs). “adj.” compares $\Delta P = P_{\text{actual}} - P_{\text{passive}}$ between conditions (Sect. 5.2).

Outcome	Comparison	Δmedian [CI]	r	p
all state outcomes (FL, HS)	B vs A	all differences $\equiv 0$	—	—
story pull, raw (HS)	F vs N	+0.0066 [0.0037, 0.0095]	0.89	1.7×10^{-6}
story pull, adj. (HS)	F vs N	+0.0022 [−0.0003, 0.0034]	0.36	0.084
story pull, adj. (FL)	F vs N	+0.0004 [−0.0016, 0.0023]	0.08	0.63
cooperation (FL)	C vs A	+0.200 [0.172, 0.239]	1.00	8.9×10^{-15}
cooperation (HS)	C vs A	+0.209 [0.170, 0.248]	0.99	9.3×10^{-9}
fragmentation (FL)	F vs A	−0.017 [−0.034, −0.006]	−0.53	9.5×10^{-4}
$\Delta\bar{\gamma}$ (HS)	F vs A	−0.336 [−0.456, −0.281]	−1.00	1.9×10^{-9}
$\Delta\bar{\gamma}$ (HS)	N vs A	−0.323 [−0.431, −0.241]	−0.94	2.0×10^{-7}
recovery time (FL, HS)	C, F, N vs A	$\Delta\text{median} = 0$ $ r < 0.5$		> 0.22

ordering. Crucially, P can also be evaluated on trajectories that never received the reference: for each treated run we compute P on the paired-seed *untreated* trajectory (A/B) scored against a matched reference of the same type, defining the causal steering effect $\Delta P = P_{\text{actual}} - P_{\text{passive}}$ (Sect. 5.2). Campaigns: a *flagship* run (conditions A/B/C/F_{invert}/N $\times$ 50 paired seeds, mild shock), a *harsh-shock* replication (removal of 40% of hubs, $\times$ 30 seeds), and a *sensitivity sweep* (feedback gain $g \in \{0.2, 0.8, 1.6\} \times \{A, C, F_{\text{invert}}, F_{\text{crisis}}, F_{\text{utopia}}, N\} \times 20$ seeds), a *scale check* re-running the core conditions with doubled population and resource capacity, a *response-regime campaign* repeating all broadcast conditions with conformist agents (15 seeds per cell), and a *replay control* (30 receiving runs paired with the harsh-shock seed list, plus 15 auxiliary source runs that generate the replayed trajectories): the reported evidence base comprises $250 + 150 + 360 + 90 + 32 + 30 + 15 = 927$ runs, each of 3,000–4,000 steps. Because conditions share seed lists, the design is paired; the primary analysis is a Wilcoxon signed-rank test on per-seed differences, reported with the matched-pairs rank-biserial correlation r and a bootstrap 95% CI on the median difference. Unpaired Mann–Whitney U with Cliff’s δ is retained as a robustness check and agrees with the paired analysis throughout; all other logged measures are exploratory.

5 Results

Table 3 summarizes the main paired comparisons; the following subsections unpack them.

5.1 Measurement alone is causally inert

Under paired seeds, condition B is trajectory-identical to A: every per-seed difference is exactly zero, on every outcome, in both the flagship and harsh-shock

campaigns. The instrument acquires causal power only when it broadcasts — a validation that the design isolates the channel of interest.

5.2 Apparent story pull is explained by a passive counterfactual

Except where stated otherwise, all results concern the primary corrective regime; Sect. 5.4 shows how they change under conformity.

Evaluated naively, the story-pull statistic appears to show strong self-fulfilling steering. Raw P (harsh shock; medians, bootstrap 95% CIs) is positive in all three non-truth signal conditions — F 0.0078 [0.0066, 0.0100], N 0.0014 [0.0001, 0.0023] — and largest for the replayed genuine self-model of a sibling world (condition R; 30 receivers on the harsh-shock seed list, one of 15 source trajectories each): 0.0241 [0.0197, 0.0265]. The raw hierarchy $R > F > N$ is highly reliable (F vs N $r = 0.89$, $p \approx 1.7 \times 10^{-6}$; robust to one receiver per source, $n = 15$, $r = 1.00$, and to a source-clustered bootstrap).

The passive counterfactual dissolves this picture. For each treated run we score the paired-seed *untreated* trajectory (A, or the identical B) against a matched reference of the same type, post hoc: the inverted mirror of its own state, the same replayed source trajectory, or the same noise sequence. Untreated worlds “move toward” these references *more* than treated worlds move toward their actual broadcasts — passive medians 0.0119 (invert), 0.0300 (replay), 0.0073 (noise) — giving uniformly negative ΔP : -0.0036 (F), -0.0058 (R), -0.0055 (N), each $p \leq 10^{-5}$. Between-condition contrasts of ΔP are not significant in either campaign (F vs N: harsh $p = 0.084$; flagship $p = 0.63$; R vs N: $p = 0.95$). Raw P is therefore dominated by intrinsic dynamics — post-shock relaxation and drift toward typical states — that align best with realistic references (the replayed trajectory shares the ecology’s attractor and the common shock time; excluding the shock transition changes medians by $< 2\%$) and worst with unstructured noise. Response activation confirms the reading: F keeps four of five corrective rules active on $\geq 90\%$ of ticks while R is mostly quiescent ($\langle |b_k - s_k| \rangle$: 0.64, 0.36, 0.10 for F, N, R), yet R shows the largest raw pull — it was never response-driven. We conclude there is *no evidence of content-specific attraction toward broadcast content*; the small negative ΔP , similar across signal types, indicates broadcasts mildly damp the intrinsic drift. This is the paper’s methodological result: a statistic of the kind natural in empirical performativity studies registers large, internally consistent, and entirely artifactual “steering” unless checked against passive twins.

Structural spillovers, by contrast, are real. Direct outcome contrasts are unambiguous: agents persistently told the network is fragmenting over-connect it (Fig. 3), reducing post-shock fragmentation (flagship paired Δ median -0.017 , $r = -0.53$, $p \approx 10^{-3}$; directionally consistent, ns, under harsh shock). Broadcasts densify and flatten the network: relative to A (paired, harsh shock, post-transient medians), mean degree rises $+1.1$ under truth ($r = 0.49$), $+3.8$ under the inverted lie ($r = 0.80$), $+2.8$ under noise ($r = 0.73$), while Freeman centralization (F: -0.026 , $r = -0.74$; N: -0.010) and betweenness concentration

Sensitivity sweep: 40% hub-removal shock, 20 seeds per cell

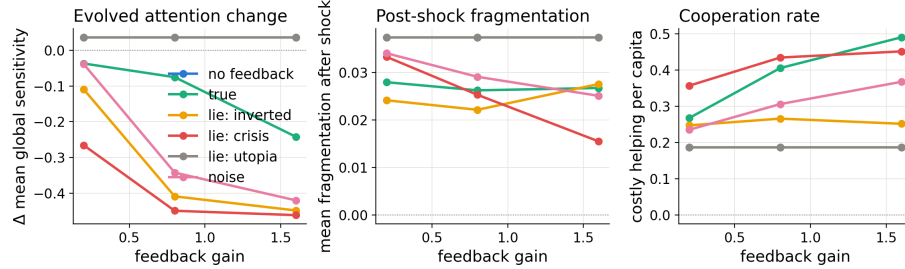


Fig. 2. Sensitivity sweep (40% hub-removal shock; 20 seeds per cell; medians). Left: evolved change in mean global sensitivity γ — attention to the self-model declines more strongly with increasing broadcast unreliability and gain; the flattering (utopia) lie tracks the no-feedback baseline exactly. Middle: post-shock fragmentation — under corrective responses the alarmist (crisis) lie does *not* come true. Right: cooperation rate — deficit signals mobilize costly helping.

(F: -0.007 , $r = -0.67$; N: -0.008) fall: the fed-back network is denser and less hub-dominated than the silent one.

5.3 The type of lie sets what changes

The three distortion modes produce qualitatively different *outcome* regimes (Fig. 2). The *inverted* self-model, always reporting the mirror of reality, keeps most corrective responses persistently active (four of five on $\geq 90\%$ of ticks): it mobilizes costly helping ($+0.106$ over baseline, flagship $r = 0.82$) and drives the over-connection above. The *crisis* self-model is self-defeating for structure — corrections push fragmentation away from the alarmist report — yet its permanent cooperation-deficit signal mobilizes the highest costly-helping rates observed under correction (0.35–0.45 across gains). The *utopia* self-model is behaviorally inert: because responses fire only on reported deficits, a flattering broadcast triggers nothing and is indistinguishable from disconnecting the channel, at every gain. Under purely corrective response rules, flattery equals silence. We note the caveat here rather than in the discussion: the first-order *directions* of these responses follow from the corrective rules (Eq. 1) by construction and are not claimed as discoveries; what is not built in is the magnitude asymmetry between distortion types, the spillovers to non-targeted structure, and the evolutionary consequences below.

5.4 The response regime decides which lies come true

Repeating all broadcast conditions with conformist agents reverses the taxonomy exactly as the mechanism predicts (Table 4). The *crisis* lie, self-defeating under correction, becomes strongly self-fulfilling under conformity: told the world

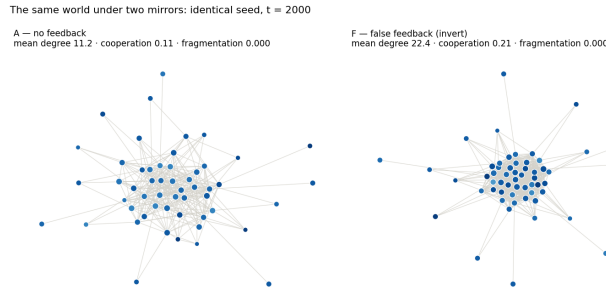


Fig. 3. The same world under two mirrors. Two runs from an identical seed at $t = 2000$: without feedback (left) and under the inverted false broadcast (right), which persistently reports fragmentation and thereby drives over-connection (mean degree 22.4 vs 11.2) and elevated costly helping. Node size encodes energy; color encodes generation.

Table 4. The same lie under two response regimes (medians; harsh shock, $g = 0.8$). Baseline (A): cooperation 0.18, fragmentation 0.038.

Lie	Corrective		Conformist	
	coop.	fragm.	coop.	fragm.
Invert	0.27	0.021	0.42	0.054
Crisis	0.36	0.025	0.26	0.400
Utopia	0.19 (inert)	0.037 (inert)	0.428	0.025

is fragmenting, imitators cut ties and fragmentation rises an order of magnitude above baseline (0.400 vs 0.038). The *utopia* lie flips from behaviorally inert to benevolently self-fulfilling: “everyone cooperates” becomes true by imitation (cooperation 0.428). Under conformity even noise mobilizes: imitating randomly reported cooperation levels raises costly helping to 0.445, as high as any lie, since imitation amplifies whatever is reported. Which false self-models come true — in the outcome sense — is therefore a property of the lie–response pair, not of the lie alone.

5.5 Populations fed lies evolve to stop listening

Mean heritable global sensitivity $\bar{\gamma}$ changes over ~ 40 generations as an emergent, unprogrammed function of broadcast quality (Fig. 4). At $g = 0.8$ (harsh-shock campaign): no broadcast -0.03 (neutral drift), truth -0.10 , noise -0.34 (paired $r = -0.94$, $p \approx 2 \times 10^{-7}$), false -0.42 (paired Δ median -0.336 $[-0.456, -0.281]$, $r = -1.00$, $p \approx 1.9 \times 10^{-9}$). The sweep reveals a dose–response: the decline under the false broadcast deepens from -0.12 ($g = 0.2$) to -0.45 ($g = 1.6$) — and at the highest gain even *accurate* feedback shows a marked decline (-0.24): high-impact feedback is disfavored even when true. The utopia lie, being consequence-free, produces no decline at all.

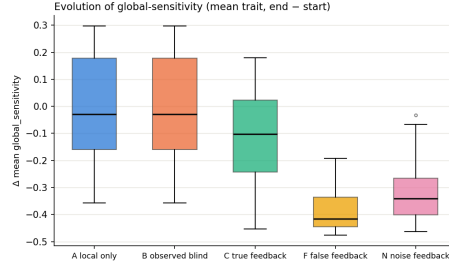


Fig. 4. Evolution of attention to the self-model (harsh shock, $g = 0.8$, 30 seeds per condition): change in mean $\bar{\gamma}$, run start to end. The ordering (false < noise < true < none) emerged from the evolutionary dynamics alone; nothing in the rules references broadcast reliability.

We state the claim at the level the data support: a systematic, dose-dependent shift in the frequency of a heritable trait under matched conditions, whose individual-level pathway is not a simple fecundity gradient (within-run $\rho \approx +0.05$ between γ_i and offspring count under F) and not shock-mediated (-0.40 without the shock vs -0.39 with it; shock victims' mean γ only $+0.03$ above the mean). The pathway plausibly runs through lineage survival and energy budgets downstream of reproduction; identifying it precisely is left open.

5.6 Accurate feedback as distributed corrective feedback

True feedback roughly doubles costly helping (flagship: 0.397 vs 0.179 ; paired Δ median $+0.200$ [$0.172, 0.239$], $r = 1.00$, $p \approx 9 \times 10^{-15}$; replicated under harsh shock, $r = 0.99$), rising monotonically with gain ($0.27 \rightarrow 0.48$). The loop implements distributed corrective feedback: deficits reported at the population level are repaired by uncoordinated individual action.

5.7 A pre-specified null result

We detected no evidence of a feedback effect on post-shock recovery time in any condition or campaign, even with 40% of hubs removed (all $p > 0.12$; median ≈ 40 steps throughout). Within the regimes tested, resilience appears demographic — fast regrowth — rather than informational; we report this pre-specified null as observed.

5.8 Robustness to system scale

Re-running the core conditions (A, C, F_{invert} , N) with doubled initial population and proportionally scaled resources (steady-state population ≈ 120 – 130 vs. ≈ 60) preserves every effect under the paired analysis ($n = 8$ seed pairs per condition): cooperation under true feedback is 0.399 vs. 0.188 baseline (Δ median

+0.151 [0.119, 0.247], $r = 1.00$, $p = 0.008$), and the distrust ordering is intact (F vs. A: Δ median -0.299 [$-0.388, -0.212$], $r = -1.00$, $p = 0.008$; N vs. A: -0.219 , $r = -0.89$, $p = 0.023$). The principal phenomena therefore persist under a $2\times$ increase in system scale.

6 Discussion and Limitations

As noted alongside the results, the first-order response directions are encoded in the rules; the findings live one level up. First, the methodological lesson: a statistic that looked like compelling evidence of self-fulfilling dynamics — positive, replicated, and largest for realistic signals — was fully explained by intrinsic dynamics on passive twin trajectories. Empirical claims that deployed predictions steer the systems they describe [19, 21] face exactly this confound, without access to the untreated twin our platform provides. Second, the substantive findings: broadcasts change behavior and structure with a regime-dependent taxonomy of lies, and above all the evolutionary layer, in which the credibility of a collective self-model becomes an evolving property of the population. Limitations: a single ecology and parameter family (though effects persist across an $8\times$ gain range and a $2\times$ scale change), fixed rule-based rather than learned response policies, and one network per run rather than interacting populations.

The results have a practical reading for engineered systems. Metric-driven infrastructure — autoscalers, observability pipelines, trending algorithms — constitutes exactly this loop; the false-broadcast conditions model telemetry corruption, and the utopia result suggests that under deficit-driven responses *flattering telemetry can be operationally equivalent to no telemetry* while raising no alarms. Documented episodes of measurement reshaping the measured system, such as ranking reactivity in institutions [4], are natural empirical targets — with the passive-counterfactual check as the entry requirement. For multi-agent AI, where populations of language-model agents are increasingly studied and deployed [15] and shared dashboards of collective state are a natural design, the evolution-of-distrust result suggests that persistent misreporting does not merely misdirect a population — it degrades the control channel itself.

7 Conclusion

We introduced a reproducible causal laboratory for reflexive self-modeling in adaptive networks and showed that (i) the measurement-only control is trajectory-identical to baseline, while broadcasting changes behavior, structure, and evolution; (ii) apparent self-fulfilling “story pull” — strong, replicated, largest for realistic signals — vanishes against a passive counterfactual, a caution for empirical performativity claims; (iii) which lies come true, in the outcome sense, is set by the lie–response pair, not the lie alone; and (iv) heritable attention to the collective self-model declines across generations, more strongly with increasing

self-model unreliability and impact, making a population’s trust in its own self-model an evolving quantity that unreliable observers spend down. Future work: learned response policies and language-model agents.

Reproducibility. Code, campaign configurations, all analysis scripts, pinned dependency versions, and deterministic regeneration of every run are public at github.com/Ivan825/Project-ONE, tag `cn2026-submission`; stored run records and replay-source trajectories ship as release assets on that tag.

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